

Learn and Improve HTML5 & CSS

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1. Goal

This guide provides a practical, beginner-friendly path to learn HTML5 and CSS, improve accessibility, and follow modern best practices — using trusted references like W3C, MDN Web Docs, and web.dev.

2. Core Learning Path (Recommended)

1. HTML5 Foundations

- Learn semantic elements: **header**, **nav**, **main**, **article**, **section**, **footer**
- Understand headings (H1–H6), lists, links, images, forms
- Practice structuring content for meaning (not appearance)

2. CSS Fundamentals

- The box model: margin, border, padding, content
- Selectors and specificity (avoid over-specific selectors)
- Layout with Flexbox and CSS Grid
- Responsive design with media queries
- Colors, typography, spacing scales

3. Modern Styling Patterns

- CSS Variables (custom properties) for design tokens
- Utility classes vs. component styles

- Organize CSS by sections/modules
- 4. Accessibility (A11y)
 - Use semantic HTML first; ARIA to fill gaps
 - Keyboard navigation and focus styles
 - Color contrast (WCAG AA), alt text for images
- 5. Performance & SEO Basics
 - Optimize images (size/format), cache headers
 - Meta tags, page titles, sitemaps, robots.txt
 - Mobile-first considerations

3. Practice: Small, Focused Exercises

- Rebuild your homepage layout using only Flexbox
- Create a two-column layout with CSS Grid and make it responsive
- Replace hardcoded values with CSS variables in a small component
- Add keyboard navigation tests (Tab through links and controls)
- Validate your pages using the W3C Validator and fix errors

4. Trusted Learning Resources

4.1. Official Standards and References

- W3C HTML Specification (Beginner Guides & Tutorials) <https://www.w3.org/standards/webdesign/htmlcss>
- W3C Web Accessibility Initiative (WAI) <https://www.w3.org/WAI/standards-guidelines/wcag/>

4.2. MDN Web Docs (Highly Recommended)

- MDN Learn Web Development (HTML/CSS/JS track) <https://developer.mozilla.org/en-US/docs/Learn>
- HTML Elements Reference <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- CSS Reference <https://developer.mozilla.org/en-US/docs/Web/CSS/Reference>
- Flexbox Guide https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Flexible_Box_Layout/Basic_Concepts_of_Flexbox
- Grid Guide https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Grid_Layout
- CSS Custom Properties (Variables) https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties

4.3. web.dev (Google)

- Responsive Design Fundamentals <https://web.dev/learn/design/>
- Performance Fundamentals <https://web.dev/learn/performance/>
- Accessibility <https://web.dev/learn/accessibility/>

4.4. Accessibility (Practical Tools)

- WAVE WebAIM Accessibility Evaluation Tool <https://wave.webaim.org/>
- Contrast Checker <https://webaim.org/resources/contrastchecker/>

5. Validate and Test Your Work

- HTML Validator <https://validator.w3.org/>
- CSS Validator <https://jigsaw.w3.org/css-validator/>
- Lighthouse (in Chrome DevTools) for performance, a11y, SEO checks

6. Suggested Project Milestones

- Milestone 1: Semantic Structure
- Convert all pages to use proper semantic tags and headings
- Milestone 2: Responsive Layout
- Ensure every page works on mobile, tablet, desktop
- Milestone 3: Theming with Variables
- Move colors/spacing to `:root` and document tokens
- Milestone 4: Accessibility Pass
- Keyboard navigation, alt text, ARIA where needed, color contrast
- Milestone 5: Performance & SEO
- Optimize images, add sitemap, check robots, meta descriptions

7. Keep Improving

- Read code by reputable open-source projects (HTML/CSS only sites)
- Practice recreating components from screenshots
- Participate in challenges (e.g., Frontend Mentor)
- Keep a personal style guide of tokens and patterns

8. Using This Project as a Learning Sandbox

- Try adding a new page and wire it into the nav
- Create an alternate theme using the variables in `css/style.css`
- Replace the gallery with a responsive Grid layout variant
- Add a print stylesheet and test printing a page
- Validate and fix any issues reported by validators and Lighthouse